# A Diagnostic Interview

## The Schedule for Affective Disorders and Schizophrenia

Jean Endicott, PhD, Robert L. Spitzer, MD

• **The Schedule for Affective Disorders and Schizophrenia (SADS) was developed to reduce information variance in both the descriptive and diagnostic evaluation of a subject. The SADS is unique among rating scales in that it provides for (1) a detailed description of the features of the current episode of illness when they were at their most severe; (2) a description of the level of severity of manifestations of major dimensions of psychopathology during the week preceding the evaluation, which can then be used as a measure of change; (3) a progression of questions and criteria, which provides information for making diagnoses; and (4) a detailed description of past psychopathology and functioning relevant to an evaluation of diagnosis, prognosis, and overall severity of disturbance. This article reports on initial scale development and reliability studies of the items and the scale scores.**

**(*Arch Gen Psychiatry* 35:837-844, 1978)**

Many evaluation procedures have been developed for obtaining reliable clinical judgments on rating scales descriptive of dimensions of psychopathology and functioning. Unfortunately, this has not been the case for diagnostic evaluations. To the extent that a procedure for making diagnostic judgments is unreliable, a limit is placed on its validity for any use. The two major sources of unreliability that lead to disagreement among clinicians on the psychiatric diagnoses given to patients are *criterion variance* and *information variance*. Criterion variance refers to differences in the inclusion and exclusion criteria that clinicians use to summarize patient data into psychiatric diagnoses. Information variance occurs when clinicians have different amounts and kinds of information about patients. With the assistance of other participants in the National Institute of Mental Health (NIMH) Clinical Research Branch Collaborative Program on the Psychobiology of Depression, we have developed several interrelated procedures that attempt to reduce greatly both criterion and information variance and thus to improve considerably the reliability, and thereby the validity, of diagnostic practice. These include the Research Diagnostic Criteria (RDC) and the Schedule for Affective Disorders and Schizophrenia (SADS).

### THE RESEARCH DIAGNOSTIC CRITERIA

In order to reduce the *criterion* variance in the diagnosis of a selected group of functional psychiatric disorders, the Research Diagnostic Criteria were developed.[1] The RDC contain specific inclusion and exclusion criteria for each of the diagnoses included. Initial reliability and validity data for the RDC are reported elsewhere.[2]

### THE SCHEDULE FOR AFFECTIVE DISORDERS AND SCHIZOPHRENIA

In conjunction with the development of the RDC, the Schedule for Affective Disorders and Schizophrenia was also developed in an effort to reduce *information* variance in both the descriptive and diagnostic evaluation of a subject. The organization of the interview and the item coverage is designed to elicit information necessary for making diagnoses using the RDC, as well as to describe the subjects' symptoms and levels of functioning.

Because of the large number of investigators who are using SADS in a variety of research studies, this article has been written to describe the various versions of the SADS and initial reliability data collected in the collaborative study.

### COVERAGE AND FORMAT OF THE SADS

There are three versions of the Schedule for Affective Disorders and Schizophrenia—the regular version (SADS),

Accepted for publication Dec 18, 1977.

From the Biometrics Research Unit, New York State Psychiatric Institute, and the Department of Psychiatry, Columbia University College of Physicians and Surgeons.

Read before the 130th annual meeting of the American Psychiatric Association, Toronto, May 1977.

Reprint requests to Biometrics Research, New York State Psychiatric Institute, 722 W 168th St, New York, NY 10032 (Dr Spitzer).



Downloaded From: http://archpsyc.jamanetwork.com/ by a University of Pittsburgh User on 09/07/2013

the lifetime version (SADS-L), and the version for measuring change (SADS-C).

Both the SADS and the SADS-L have a number of features that are not found in other evaluation procedures. The SADS is organized into two parts. Part 1 is designed to yield an extremely detailed description of the subject's current episode or condition as well as his functioning during the week prior to the interview. Part 2 is primarily for describing past psychiatric disturbance. The SADS-L is similar to part 2 of the SADS except that the time period is not limited to the past and includes any current disturbance. The SADS-L is thus more suitable for use in studies in which there is no current episode of illness, eg, with relatives of psychiatric patients, with outpatients, or where detailed information about the phenomenology of the current condition is not needed.

Part 1 of the regular SADS is unique among rating scale procedures in that the items are used to describe the features of the current episode of illness *when they were at their most severe.* This is in contrast to most psychiatric rating scales, which cover a set time period regardless of the timing of the evaluation in relation to the beginning and peak of the episode. In addition to the collection of data relevant to an evaluation of the diagnosis, prognosis, and phenomenology of a current episode of illness, Part 1 of the SADS is also designed to be appropriate for use as a measure of change. Many of the scaled items are also judged for the level of severity during the week preceding the evaluation as well as for each symptom when it was at its most severe during the current episode. This same subset of items can then be used for subsequent evaluations to measure change using the change version of the SADS (SADS-C).

The organization of the SADS and SADS-L is similar to that of a clinical interview focused on differential diagnosis. The schedule provides for a progression of questions, items, and criteria that systematically rule in and rule out specific RDC diagnoses. To give the reader a sense of the coverage of the SADS, Table 1 lists the RDC diagnoses considered during the course of the interview.

Figure 1 shows several items for part 1 (current section) of the SADS. As can be seen, there are defined levels of severity for each item, rather than exclusive reliance on the rater's use of terms such as mild, moderate, or severe. In addition to following the interview guide, the interviewer is instructed to use all sources of information and as many general or specific questions as necessary to obtain the information required to make judgments on the items. Figure 2, a page from part 2 (past section) of the SADS, illustrates the way in which the criteria for a given diagnosis are specified and the manner in which they are applied during the interview. The specific criteria in the RDC and the items of the SADS usually refer to symptoms, duration or course of illness, or severity of impairment.

Table 1.—Research Diagnostic Criteria Diagnoses Covered by the Schedule for Affective Disorders and Schizophrenia Interview

| |
|---|
| Schizophrenia<br>Acute—chronic<br>Paranoid<br>Disorganized<br>Catatonic<br>Mixed (undifferentiated)<br>Residual |
| Schizo-affective disorder—manic<br>Acute—chronic<br>Mainly schizophrenic<br>Mainly affective |
| Schizo-affective disorder—depressed<br>Acute—chronic<br>Mainly schizophrenic<br>Mainly affective |
| Depressive syndrome superimposed on residual schizophrenia |
| Manic disorder |
| Hypomanic disorder |
| Bipolar with mania (bipolar I)* |
| Bipolar with hypomania (bipolar II)* |
| Major depressive disorder<br>Primary<br>Secondary<br>Recurrent unipolar*<br>Psychotic<br>Incapacitating<br>Endogenous<br>Agitated<br>Retarded<br>Situational<br>Simple<br>Predominant mood |
| Minor depressive disorder with significant anxiety |
| Intermittent depressive disorder* |
| Panic disorder |
| Generalized anxiety disorder with significant depression |
| Cyclothymic personality* |
| Labile personality* |
| Briquet's disorder (somatization disorder)* |
| Antisocial personality* |
| Alcoholism |
| Drug use disorder |
| Obsessive compulsive disorder |
| Phobic disorder |
| Unspecified functional disorder |
| Other psychiatric disorder |
| Schizotypal features* |
| Currently not mentally ill |
| Never mentally ill* |

*These conditions are diagnosed on a longitudinal or lifetime basis. All other conditions are diagnosed on the basis of current or past episodes of psychopathology.

## PERSONNEL AND TRAINING

The most suitable personnel for administering the SADS and the SADS-L, and for using the RDC, are individuals with experience in interviewing and making judgments about manifest psychopathology. Although questions are provided to assist in collecting the information and all of the items are defined to assure uniform criteria for all raters, the types of judgments called for require more knowledge of psychiatric concepts than do many of the more commonly used psychiatric rating scales. For this reason, raters should usually be limited to psychiatrists, clinical psychologists, or psychiatric social workers. If other research personnel are to be used, much more training is generally necessary. An experienced and well-trained interviewer who is familiar with the SADS and the RDC should usually be able to complete an evaluation in 1½ to 2 hours depending on the degree of disturbance of the



Downloaded From: http://archpsyc.jamanetwork.com/ by a University of Pittsburgh User on 09/07/2013

SCREENING ITEMS FOR MANIC SYNDROME

The next 5 items are screening items to determine the presence of manic-like behavior. If any of the items are judged present, inquire in a general way to determine how he was behaving at that time with such questions as, *When you were this way, what kinds of things were you doing? How did you spend your time?* Do not include behavior which is clearly explainable by alcohol or drug intoxication.

If the subject has only described dysphoric mood, the following questions regarding the manic syndrome should be introduced with a statement such as: *I know you have been feeling (depressed). However, many people have other feelings mixed in or at different times so it is important that I ask you about those feelings also.*

Elevated mood and/or optimistic attitude toward the future which lasted at least several hours and was out of proportion to the circumstances.

*Have (there been times when) you felt very good or too cheerful or high —not just your normal self?*

If unclear: *When you felt on top of the world or as if there was nothing you couldn't do?*

*(Have you felt that everything would work out just the way you wanted?)*

*If people saw you would they think you were just in a good mood or something more than that?*

*(What about during the past week?)*

0 No information
1 Not at all, normal, or depressed
2 Slight, e.g., good spirits, more cheerful than most people in his circumstances, but of only possible clinical significance
3 Mild, e.g., definitely elevated mood and optimistic outlook that is somewhat out of proportion to his circumstances
4 Moderate, e.g., mood and outlook are clearly out of proportion to circumstances
5 Severe, e.g., quality of euphoric mood
6 Extreme, e.g., clearly elated, exalted expression and says "Everything is beautiful, I feel so good"

353

PAST WEEK 0 1 2 3 4 5 6 354

Less need for sleep than usual to feel rested (average for several days when needed less sleep).

*Have you needed less sleep than usual to feel rested? (How much sleep do you ordinarily need?) (How much when you were [are] high?)*

*(What about during the past week?)*

0 No information
1 No change or more sleep needed
2 Up to 1 hour less than usual
3 Up to 2 hours less than usual
4 Up to 3 hours less than usual
5 Up to 4 hours less than usual
6 4 or more hours less than usual

355

PAST WEEK 0 1 2 3 4 5 6 356

Unusually energetic, more active than his usual level without expected fatigue.

*Have you had more energy than usual to do things?*

*(More than just a return to normal or usual level?)*

*(Did it seem like too much energy?)*

*(What about during the past week?)*

0 No information
1 No different than usual or less energetic
2 Slightly more energetic but of questionable significance
3 Little change in activity level but less fatigued than usual
4 Somewhat more active than usual with little or no fatigue
5 Much more active than usual with little or no fatigue
6 Unusually active all day long with little or no fatigue

357

PAST WEEK 0 1 2 3 4 5 6 358

Fig 1.—Page 19 from part 1 of SADS.



Downloaded From: http://archpsyc.jamanetwork.com/ by a University of Pittsburgh User on 09/07/2013

CRITERIA FOR EPISODES OF MANIC SYNDROME (There are 3 criteria.)

I Has had 1 or more distinct periods lasting at least 1 week (or any duration if hospitalized) when the predominant mood was either elevated (i.e., unusually good, cheerful, high, expansive) OR irritable (i.e., easily annoyed). (Do not include if apparently due to alcohol or drug intoxication.) Note: This is frequently falsely rated as positive when the subject is merely describing feeling very good in contrast to periods of depression.

*Did you ever have a period that lasted at least a week (or when you were hospitalized), when you felt extremely good or high —clearly different from your normal self? Did friends or your family think that this was more than just feeling good?*

*What about periods when you felt very irritable and easily annoyed?*

[0] No information or not sure 550
[1] Never had a period lasting at least 1 week (or was hospitalized) when the predominant mood was either elevated or irritable
2 Never had a period when predominant mood was elevated, but had at least 1 period when he was irritable
3 Had at least 1 period when predominant mood was elevated (with or without irritability)

→ Skip to Episodes of Major Depressive Syndrome, page 47.

II Had at least 2 symptoms associated with the most severe period of euphoric or irritable mood (inquire for all symptoms). (Do not include if apparently due to alcohol or drug intoxication.)

| | DURING THE MOST SEVERE PERIOD No Info | No | Yes | |
|---|---|---|---|---|
| *During the most severe period.....* | | | | |
| *...were you more active than usual —either socially, at work, sexually, or physically restless?* | X | 1 | 2 | 551 |
| *...were you more talkative than usual or felt a pressure to keep on talking?* | X | 1 | 2 | 552 |
| *...did your thoughts race or did you talk so fast that it was difficult for people to follow what you were saying?* | X | 1 | 2 | 553 |
| *...did you feel you were a very important person, had special plans, powers, talents, or abilities (grandiosity)?* | X | 1 | 2 | 554 |
| *...did you need less sleep than usual?* | X | 1 | 2 | 555 |
| *...did you have trouble concentrating on what was going on because your attention kept jumping to unimportant things around you (distractibility)?* | X | 1 | 2 | 556 |
| *...did you do anything foolish that could have gotten you into a lot of trouble, like buying things, business investments, sexual indiscretions, reckless driving?* | X | 1 | 2 | 557 |

Number of definite symptoms = ___. If euphoric, criterion = 2; if irritable only, criterion = 3. 558

☐ If criterion II is not met, check here and skip to Episodes of Major Depressive Syndrome, page 47. 559

III Symptoms were so severe that meaningful conversation was impossible, there was serious impairment in functioning, or he was hospitalized.

*Were you hospitalized? Were you so excited that it was almost impossible to hold a conversation with you? Did it cause troubles with people, with your family your work, or other usual activities?*

[0] No information 560
[1] No
2 Yes

→ Skip to Episodes of Major Depressive Syndrome, page 47.

Fig 2.—Page 43 from part 2 of SADS.



Downloaded From: http://archpsyc.jamanetwork.com/ by a University of Pittsburgh User on 09/07/2013

Table 2.—Item Content of Summary Scale Scores of the Current Section of the Second Edition of the Schedule for Affective Disorders and Schizophrenia

| No. | Item Name | Weight |
|---|---|---|
| **Scale 1—Depressive Mood and Ideation** | | |
| 234 | Depressive mood | 1-7 |
| 238 | Brooding | 1-6 |
| 240 | Feelings of self-reproach | 1-6 |
| 242 | Negative evaluation of self | 1-6 |
| 244 | Discouragement | 1-6 |
| | | (5-31) |
| **Scale 2—Endogenous Features** | | |
| 236 | Distinct quality of mood | 1-4 |
| 237 | Lack of specific concern | 1-4 |
| 240 | Feelings of self-reproach | 1-6 |
| 272 | Insomnia | 1-6 |
| 274 | Middle insomnia | checklist |
| 275 | Terminal insomnia | checklist |
| 317 | Loss of appetite | 1-6 |
| 319 | Weight loss | 1-6 |
| 326 | Loss of interest or pleasure | 1-6 |
| 334 | Psychomotor agitation | 1-6 |
| 342 | Psychomotor retardation | 1-6 |
| 350 | Reactivity | 1-6 |
| 351 | Worse in morning | 1-4 |
| | | (11-62) |
| **Scale 3—Depressive—Associated Features** | | |
| 246 | Suicidal tendencies | 1-7 |
| 272 | Insomnia | 1-6 |
| 273 | Initial insomnia | checklist |
| 274 | Middle insomnia | checklist |
| 275 | Terminal insomnia | checklist |
| 314 | Sleeps more | 1-6 |
| 315 | Lack of energy | 1-6 |
| 317 | Loss of appetite | 1-6 |
| 319 | Weight loss | 1-6 |
| 320 | Appetite increase | 1-6 |
| 321 | Weight gain | 1-6 |
| 324 | Indecisiveness | 1-6 |
| 325 | Difficulty concentrating | 1-6 |
| 326 | Loss of interest or pleasure | 1-6 |
| 334 | Psychomotor agitation | 1-6 |
| 342 | Psychomotor retardation | 1-6 |
| 473 | Depressed appearance | 1-6 |
| | | (14-88) |
| **Scale 4—Suicidal Ideation and Behavior** | | |
| 246 | Suicidal tendencies | 1-7 |
| 248 | Number of attempts | 1-9 |
| 249 | Seriousness of intent | 1-6 |
| 250 | Medical threat to life | 1-6 |
| | | (4-28) |
| **Scale 5—Anxiety** | | |
| 251 | Panic attacks | 1-3 |
| 262 | Number of attacks | 1-9 |
| 263 | Somatic anxiety | 1-6 |
| 265 | Psychic anxiety | 1-6 |
| 267 | Phobias | 1-6 |
| | | (5-30) |
| **Scale 6—Manic Syndrome** | | |
| 353 | Elevated mood | 1-6 |
| 355 | Less need for sleep | 1-6 |
| 357 | Unusually energetic | 1-6 |
| 359 | Increased activity | 1-6 |
| 361 | Grandiosity | 1-6 |
| 364 | Irritability when manic | 1-6 |
| 365 | Hyperactivity when manic | 1-6 |
| 366 | Accelerated speech when manic | 1-6 |
| 367 | Speeded thinking when manic | 1-3 |
| 368 | Intrusive when manic | checklist |
| 369 | Public display | checklist |
| 370 | Financial indiscretion | checklist |
| 371 | Assumes responsibility | checklist |
| 372 | Antisocial when manic | checklist |
| 373 | Sexual excesses | checklist |
| 374 | Drunk when manic | checklist |
| 375 | Grossly bizarre behavior when manic | checklist |
| 413 | Poor judgment | 1-6 |
| | | (10-65) |
| **Scale 7—Delusions—Hallucinations** | | |
| 422 | Delusions of reference | 1-3 |
| 423 | Delusions of being controlled | 1-3 |
| 424 | Delusions-read mind | 1-3 |
| 425 | Thought broadcasting | 1-3 |
| 428 | Persecutory delusions | 1-3 |
| 430 | Delusions of guilt or sin | 1-3 |
| 431 | Grandiose delusions | 1-3 |
| 432 | Somatic delusions | 1-3 |
| 433 | Severity of delusions | 1-6 |
| 438 | Bizarreness of delusions | 1-6 |
| 439 | Multiple delusions | 1-3 |
| 442 | Auditory hallucinations | 1-3 |
| 443 | Voice commentary | 1-3 |
| 444 | Two or more voices converse | 1-3 |
| 445 | Nonaffective verbal hallucinations | 1-3 |
| 446 | Visual hallucinations—formed | 1-3 |
| 447 | Olfactory hallucinations | 1-3 |
| 448 | Somatic hallucinations | 1-3 |
| 449 | Hallucinations—grandiose or persecutory | 1-3 |
| 450 | Severity of hallucinations | 1-6 |
| | | (20-69) |
| **Scale 8—Formal Thought Disorder** | | |
| 467 | Flight of ideas | 1-6 |
| 474 | Incoherence | 1-6 |
| 475 | Loosening of associations | 1-6 |
| 513 | Illogical thinking | 1-6 |
| 514 | Poverty of content of speech | 1-6 |
| 515 | Neologisms | 1-3 |
| | | (6-33) |

subject. The SADS-L takes less time unless a subject has a very complicated psychiatric history.

## FIELD TESTING AND INITIAL USE OF THE SADS

The RDC, the SADS, and the SADS-L have undergone a number of revisions over the past four years. In addition to work by the authors and their staff, the procedures have been field-tested in two collaborative studies. The first involved 104 inpatients at five of the facilities participating in the Collaborative Program on the Psychobiology of Depression. The experience gained in that study resulted in additional revisions. The SADS has recently been used in a pilot study of 150 inpatients by four of the participating groups (Boston, Iowa City, New York, and St Louis).

## SUMMARY SCALE SCORES FOR THE CURRENT SECTION OF THE SADS

There are a number of items in the current section of the SADS that are descriptive of specific dimensions of psychopathology. An initial scoring system has been devel-



Downloaded From: http://archpsyc.jamanetwork.com/ by a University of Pittsburgh User on 09/07/2013

oped through the assignment of items to eight larger summary dimensions. The item assignment was based on knowledge of factor analytic work of scales with similar item content, consideration of the major clinical distinctions usually made in research studies of affective and schizophrenic disorders, and a desire to have a smaller number of clinically meaningful measures on which to compare individual subjects over time or groups of subjects with each other. The names of the summary scales and the item contents are shown in Table 2. The two syndromal scales, which have considerable overlap, were developed to describe features of the depressive syndrome that are frequently associated with depressive mood. The first, Endogenous Features, is limited to those items descriptive of symptoms that have traditionally been considered characteristic of an "endogenous" depressive episode. The other scale, Depressive Associated Features, is broader and includes both the "endogenous" items and some additional items generally assumed to be part of the depressive syndrome.

Further scale development will be done as data accumulates.

## RELIABILITY

The reliability of an instrument and its scoring system can be tested in a number of ways and at a number of different levels. The most common procedure for determining reliability is to calculate the intraclass correlation coefficients of reliability based on joint, but independent, evaluations made by two raters who observe the same interview. A much less common procedure for assessing the reliability of ratings of psychopathology and functioning is to have two raters independently interview the subject on different occasions. This is referred to as test-retest reliability. Coefficients of reliability for test-retest evaluations are usually considerably lower than that for joint interviews since additional sources of unreliability are operating with such a procedure. The first is subject variance, which refers to patients being in different stages of the condition on two different occasions. For example, a patient may begin hallucinating or develop delirium tremens between the first and second interview. In addition, information variance also tends to lower the test-retest reliability of a procedure since the patient may tell one rater something he fails to tell the other or may display a behavior in one setting that is not shown in the other.

A third, and less crucial, test of reliability is that of internal consistency. A measure of internal consistency indicates the degree to which the individual items within a composite summary scale covary and are related to the total score on that scale. The higher the index of internal consistency, the more certain the developer of the scale will be that the scale is a measure of a single dimension. Low internal consistency constrains validity. A scale with a relatively low index of internal consistency may still have some valid items as a measure, but the different dimensions tapped by the scale will be less clear.

The intraclass correlation coefficients of interrater reliability were calculated for the individual scaled items of the current section of the SADS as well as for the summary scale scores using data from the two studies that were conducted at the four participating facilities. In the first study, a total of 150 newly admitted inpatients who had a depressive or manic syndrome were jointly evaluated by pairs of raters. In the second study, 60 newly admitted inpatients who agreed to be interviewed twice were independently interviewed. Although the design called for the second interview to be within 24 to 48 hours of the first, 21 patients were interviewed between 48 and 72 hours and three patients between 72 hours and one week.

Table 3.—Coefficients of Reliability for Summary Scales of the Schedule for Affective Disorders and Schizophrenia

| Summary Scales | Intraclass *r* | | |
|---|---|---|---|
| | Joint Interview (N = 150) | Test–Retest (N = 60) | Internal Consistency* (N = 150) |
| Depressive mood and ideation | .95 | .78 | .87 |
| Endogenous features | .96 | .83 | .80 |
| Depressive–associated features | .96 | .88 | .79 |
| Suicidal ideation and behavior | .97 | .83 | .80 |
| Anxiety | .94 | .67 | .58 |
| Manic syndrome | .99 | .93 | .97 |
| Delusions–hallucinations | .97 | .91 | .87 |
| Formal thought disorder | .82 | .49 | .47 |

*Cronbach's Alpha.

The intraclass correlation coefficients of reliability for the 120 scaled items of the current section of the SADS were calculated. The cumulative frequencies for both joint and test-retest evaluations indicate that there were very high levels of agreement for almost all of the scaled items, with 90% of the coefficients being .60 or better for the joint evaluations and 82% being .60 or better for the test-retest evaluations. The corresponding percentages for .70 or better are 83% and 73%. When the data were examined for each of the participating facilities separately, the levels of agreement were similar. An examination of the items that were relatively less reliable under the test-retest conditions showed that, in most cases, these were items that were rarely reported on either occasion. For the most part, the items with the lowest test-retest reliability were not key diagnostic items.

The intraclass correlation coefficients of reliability for the summary scale scores are shown in Table 3 for both the joint and the test-retest evaluations. The reliabilities for the composite scales were extremely high for the joint interviews. With the exception of the Formal Thought Disorder Scale, the reliabilities of the composite scores were also quite high under the test-retest conditions. An examination of the ratings indicated that the lack of reliability of the Formal Thought Disorder Scale was primarily due to a lack of variability, in that none of the subjects were very disturbed in that area. This is not surprising since they had to be sufficiently intact to agree to being interviewed twice. We would expect that the scale would be more reliable in subjects with greater variability.



Downloaded From: http://archpsyc.jamanetwork.com/ by a University of Pittsburgh User on 09/07/2013

Table 4.—Intercorrelation Coefficients of Summary Scales of the Current Section of the Schedule for Affective Disorders and Schizophrenia*

| | Depressive Mood & Ideation | Endogenous Features | Depressive–Associated Features | Suicidal Ideation & Behavior | Anxiety | Manic Syndrome | Delusions–Hallucinations | Formal Thought Disorder |
|---|---|---|---|---|---|---|---|---|
| Depressive mood & ideation | ... | .72 | .74 | .51 | .39 | −.55 | −.15 | −.48 |
| Endogenous features | ... | ... | .91 | .40 | .44 | −.53 | −.18 | −.49 |
| Depressive–associated features | ... | ... | ... | .52 | .47 | −.54 | −.12 | −.50 |
| Suicidal ideation & behavior | ... | ... | ... | ... | .20 | −.41 | −.16 | −.34 |
| Anxiety | ... | ... | ... | ... | ... | −.22 | .09 | −.29 |
| Manic syndrome | ... | ... | ... | ... | ... | ... | .08 | .63 |
| Delusions–hallucinations | ... | ... | ... | ... | ... | ... | ... | .18 |
| Formal thought disorder | ... | ... | ... | ... | ... | ... | ... | ... |

*Based on 150 inpatients of pilot study.

Table 5.—Correlations Between SADS Summary Scale Scores and Anticipated Correlates Derived From Other Measures*

| SADS | KAS-R† (N = 50) | | KAS-S2 & 3‡ (N = 146) | | SCL-90§ (N = 144) | |
|---|---|---|---|---|---|---|
| Depressive mood and ideation | Withdrawn depression | .53 | Dissatisfaction | .37 | Depression | .68 |
| Endogenous features | Withdrawal-retardation | .53 | Dissatisfaction | .42 | Depression | .61 |
| | Withdrawn depression | .58 | | | | |
| Depressive–associated features | Withdrawal-retardation | .49 | Dissatisfaction | .46 | Depression | .62 |
| | Withdrawn depression | .51 | | | | |
| Suicidal ideation and behavior | Withdrawn depression | .42 | | | Depression | .47 |
| Anxiety | Helplessness | .40 | | | Somatization | .51 |
| | Anxiety | .28 | | | Anxiety | .56 |
| | Somatic | .53 | | | Phobic anxiety | .48 |
| Manic syndrome | Verbal expansiveness | .27 | Reported activities | .34 | | |
| | Hyperactivity | .23 | | | | |
| Delusions–hallucinations | Bizarreness | .32 | | | Paranoid ideation | .15 |
| | Acute psychoticism | .25 | | | Psychoticism | .24 |

*The anticipated correlates were specified in advance on the basis of item coverage.
†Katz Adjustment Scales completed by a relative.
‡Katz Adjustment Scales completed by the subject.
§SCL-90 completed by the subject.

Table 3 also lists the internal consistency index of reliability for the summary scales using the data from the interviewers' ratings of 150 subjects. The low internal consistency for Formal Thought Disorder is undoubtedly partially due to the relatively low variability in the sample of subjects. However, it is also likely that the items of this summary scale do not tap a unidimensional concept. An effort has been made to improve the reliability of the scale, both for test-retest reliability and for internal consistency, by clarifying the definitions. The degree to which this has been achieved awaits further testing.

## INTERCORRELATIONS AMONG THE SUMMARY SCALES

The intercorrelations among the summary scales, computed for the 150 patients described previously, are shown in Table 4. The overlap in item content accounts for the high correlation between scores on Endogenous Features and Depressive Associated Features. With the exception of these two summary scales, the intercorrelations between the scales are of moderate (or even negative) size, indicating that the dimensions described by the scales are relatively independent. The next highest correlations, those between Depressed Mood and Ideation and the two scales describing Endogenous and Associated Features, although high, indicate that little more than half of the variability in scores is shared.

## RELATIONSHIP BETWEEN SUMMARY SCALE SCORES AND OTHER MEASURES

The correlations between the SADS summary scale scores and measures derived from the SCL-90 and the KAS-R are shown in Table 5. The SCL-90 was completed by 144 of the 150 patients and 146 of them completed the KAS-S2 and 3 scales. Fifty relatives completed the KAS-R prior to the analysis of the data. Scores were selected that appeared, on the basis of content, to be tapping the same dimensions as those of the SADS summary scale scores. Although frequently low, the levels of agreement found between these relatively independent measures is as high as, and at times higher than, those usually found when multiple measures based on different sources of information are compared.[3,4]

## RELIABILITY OF DIAGNOSES BASED ON SADS AND SADS-L INTERVIEWS

One of the major purposes of the SADS and SADS-L interviews is to reduce the information and criterion variance that accounts for much of the unreliability of clinical diagnosis. To what degree is this goal achieved? The interjudge reliability of RDC diagnoses has been compared for the joint interviews and for test-retest evaluations described previously. The levels of reliability are reported in detail in another article describing the RDC.[2] The kappa coefficients of reliability found in these



Downloaded From: http://archpsyc.jamanetwork.com/ by a University of Pittsburgh User on 09/07/2013

studies have been far higher than those usually reported in studies of diagnostic reliability. This is true for both current and past episodes of illness. We have no data available that could be used to assess the degree to which the use of the SADS interview increases the reliability of the RDC diagnoses over that that would be obtained using the RDC with a nonstandardized clinical interview.

## COMMENT

The SADS and SADS-L are new procedures that have several unique features. The interview guide and items of the SADS provide for a detailed description of the current episode of illness as well as of the subject's level of psychopathology and functioning during the week prior to the evaluation. The SADS includes the specific inclusion and exclusion criteria for the selected group of mental disorders described in the Research Diagnostic Criteria. A large number of items related to prognosis and evaluation of severity of both past and present episodes of mental disorder are also included. The SADS-L has the same coverage as the SADS with the exception that it does not provide for the detailed description of the current episode.

There are two other interview schedules that have been developed with a focus on psychiatric diagnosis. The Current and Past Psychopathology Scales (CAPPS), also developed by us,[5] was designed to collect information relevant to making a group of DSM-II diagnoses.[6] The diagnoses were available only after the data were analyzed by a computer program, DIAGNO II. The logic of the decision process is not reflected in the organization of the instrument, as it is in the SADS, and the computer algorithms utilize complicated indices that can not be easily translated into rules that are clinically understandable. Furthermore, the data for the patient's current condition are limited to the last month, rather than the entire current episode of illness, so that often diagnostically significant behaviors are missed.

The Present State Examination (PSE), developed by Wing and his colleagues,[7] has been widely used in diagnostic studies. It differs from the SADS in that it does not collect historical information. Therefore, diagnoses requiring such information cannot be made, as for example, bipolar affective disorder and antisocial personality. Furthermore, the coverage does not include alcohol or drug abuse, further restricting the possible diagnoses. Finally, as with the CAPPS, the diagnoses based on the PSE information require the use of a computer program, CATEGO,[8] that also uses indices in its algorithms, which make the diagnostic rules difficult to translate into clinical terms.

In contrast to the CAPPS and the PSE, a diagnostician can make standardized diagnoses using the RDC criteria immediately after he has completed his SADS evaluation. The RDC criteria, as seen in Fig 2, are easily understood and are stated in clinical terms.

Both the items and the summary scales of the SADS have been found to have high interjudge reliability for test-retest evaluations as well as for joint interviews. There is also evidence that the use of the SADS and RDC improves the reliability of research diagnoses over that previously reported. The degree to which the increased reliability of diagnoses is based on the reduction of information variance is not known.

The SADS, SADS-L, and RDC are being used in a large number of studies to investigate a wide variety of research questions having to do with the genetics, psychobiology, response to treatment, and clinical description of different mental disorders. In addition to the participants in the collaborative program, the SADS is currently in use by a number of research investigators in North America and in Europe. Although there was little information available on the reliability and validity of the procedures, these research groups decided to use the SADS (and the RDC) because of their coverage, organization, specification of diagnostic criteria, and defined levels of severity. Furthermore, investigators who are now using the SADS and RDC know that they will be able to relate their findings to many of the diagnostic categories that are to be included in the third edition of the American Psychiatric Association's *Diagnostic and Statistical Manual* (DSM-III). DSM-III will contain operational criteria for all of its diagnostic categories, many of which will be virtually identical with or slight modifications of those contained in the RDC.

Within the next few years, evidence of the value of the SADS and RDC for different purposes will be available. To some extent, current users of these procedures are assuming that they are at least as valid as many of the previously developed assessment procedures and that they have the potential for having even greater usefulness, particularly for studies focusing on affective disorders and schizophrenic disorders.

This schedule was developed with the assistance of the other participants in the NIMH Clinical Research Branch Collaborative Program on the Psychobiology of Depression. This investigation was supported by grants MH-21411 and MH-23864 from the NIMH and by the New York State Department of Mental Hygiene.

## References

1. Spitzer RL, Endicott J, Robins E: *Research Diagnostic Criteria (RDC)*. New York, Biometrics Research, New York State Psychiatric Institute, 1975.

2. Spitzer RL, Endicott J, Robins E: Research Diagnostic Criteria: Rationale and reliability. *Arch Gen Psychiatry* 35:773-782, 1978.

3. Spitzer RL, Endicott J, Fleiss J, et al: The Psychiatric Status Schedule: A technique for evaluating psychopathology and impairment in role functioning. *Arch Gen Psychiatry* 23:41-55, 1970.

4. Ellsworth RB, Foster L, Childers B: Hospital and community adjustment as perceived by psychiatric patients, their families, and staff. *J Consult Clin Psychiatry* 32:1-41, 1968.

5. Endicott J, Spitzer RL: Current and Past Psychopathology Scales (CAPPS). *Arch Gen Psychiatry* 27:678-687, 1972.

6. Spitzer RL, Endicott J: DIAGNO II: Further developments in a computer program for psychiatric diagnosis. *Am J Psychiatry* 125:12-21, 1969.

7. Wing JK, Birley JLT, Cooper JE, et al: Reliability of a procedure for measuring and classifying (present psychiatric state). *Br J Psychiatry* 113:499-515, 1967.

8. Wing JK, Cooper JE, Sartorius N: *Description and Classification of Psychiatric Symptoms*. Cambridge, England, Cambridge University Press, 1974.



Downloaded From: http://archpsyc.jamanetwork.com/ by a University of Pittsburgh User on 09/07/2013